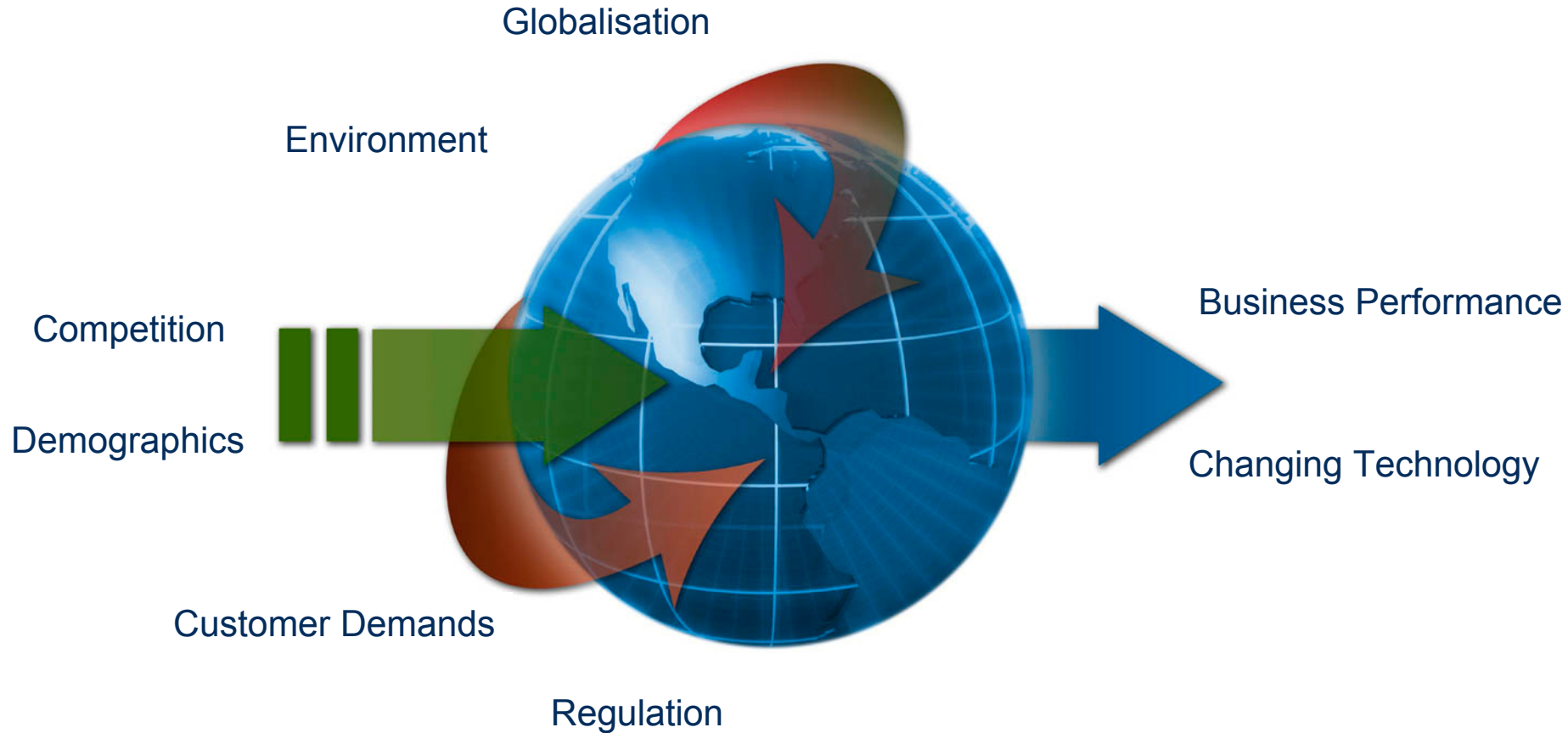


Solutions Through Partnership

Warren East
CEO

The World is Changing for Our Industry



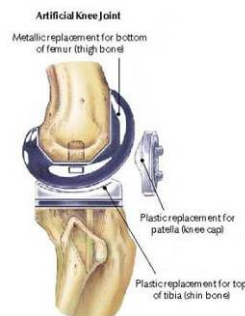
We need to evolve to meet these changes

Older and Wealthier Than Ever

- People are living longer and have more disposable income
 - Want to stay healthier longer



Keizo Mura Skiing
on his 100th Birthday



Active wrist, elbow,
knee joints to increase strength



Replacement senses

- Want to spend more on the (great) grandchildren



Vigour of Growth Economies



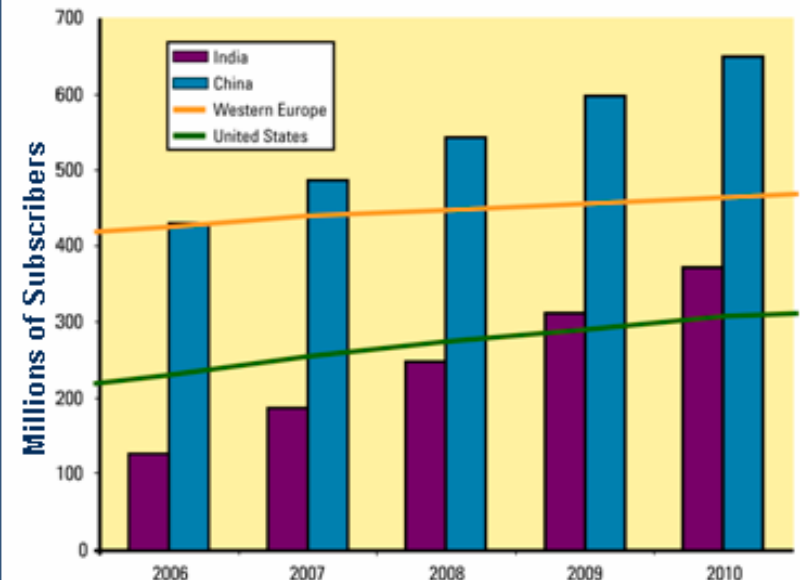
Shanghai and Beijing offices - Opened 2002

- 25 ARM processor licensees already receiving royalties
- Also libraries, fabric IP, software kits, etc.
- Strong ecosystem – 60+ books
19 in ARM Connected Community
100+ university courses

Bangalore office - Opened 2004

- 2 ARM processor licensees
- Many software development kits
- ~ 30% revenue growth
- 5 ARM Connected Community members
- 120M ARM processor-based devices sold in India in 2005

Mobile Subscriber Growth, 2006-2010



Changing Technology – Pervasiveness



1913 Model T Ford
No electronics

2006 Chrysler announce
40% of models will offer
iPod integration

Circa 1980 BMW 733i
Introduction of ABS



2005 BMW 7 Series
iDrive Control system



2015 - Pervasive
but hidden electronics



INCREASING ELECTRONIC PRESENCE

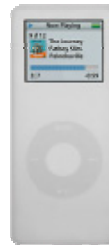
Changing Technology – Miniaturisation

- Digitisation allows increased miniaturisation, performance and convenience

The best of
analogue...



... has become
smaller and
better



Changing Technology - Performance

- Increased performance and functional integration
 - Multiple processors
 - Multiple software layers
 - Very high gate counts
 - Mixed signal
 - Platform cost put at \$1B*

*Gartner, February 2006

ARM CPU Estimator (ACE)

Device ▲▼	Name ▲▼	CPU Family ▲▼	Speed ▲▼	
	O2	Xda Exec	ARM9	520 MHz
	Dell	Axim X51v	ARM9	416 MHz
	T-Mobile	MDA III	ARM9	377 MHz
	Orange	SPV M2000	ARM9	377 MHz
	T-Mobile	MDA II	ARM9	299 MHz
	Motorola	E680	ARM9	286 MHz

www.arm.com/ace/jbench/login

ARM7



ARM9



ARM11



ARM7



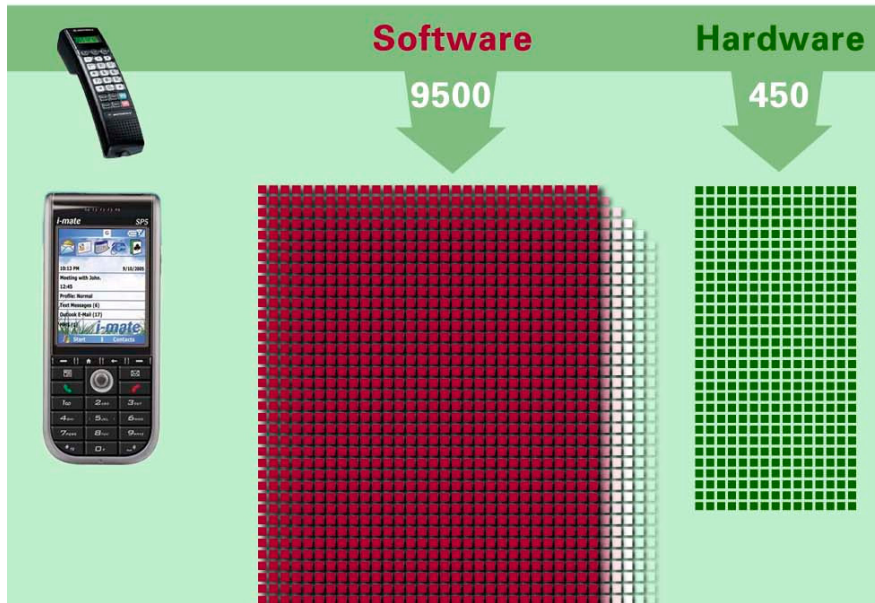
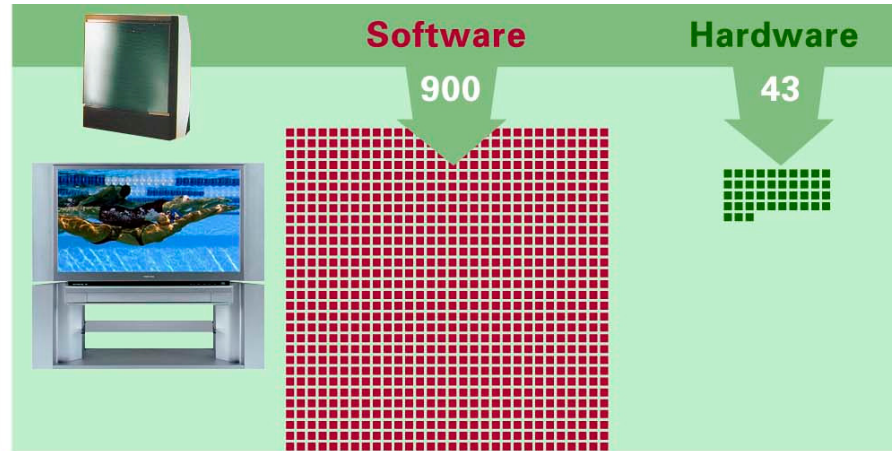
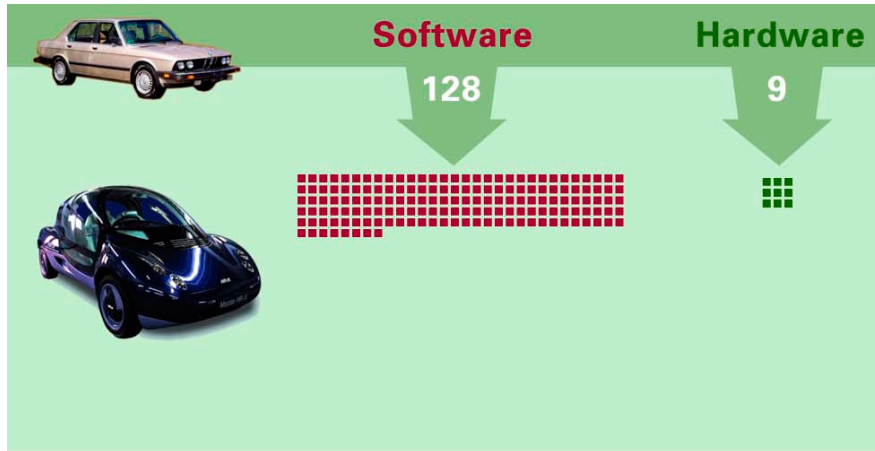
ARM9



ARM11



Increasing Software Integration Cost



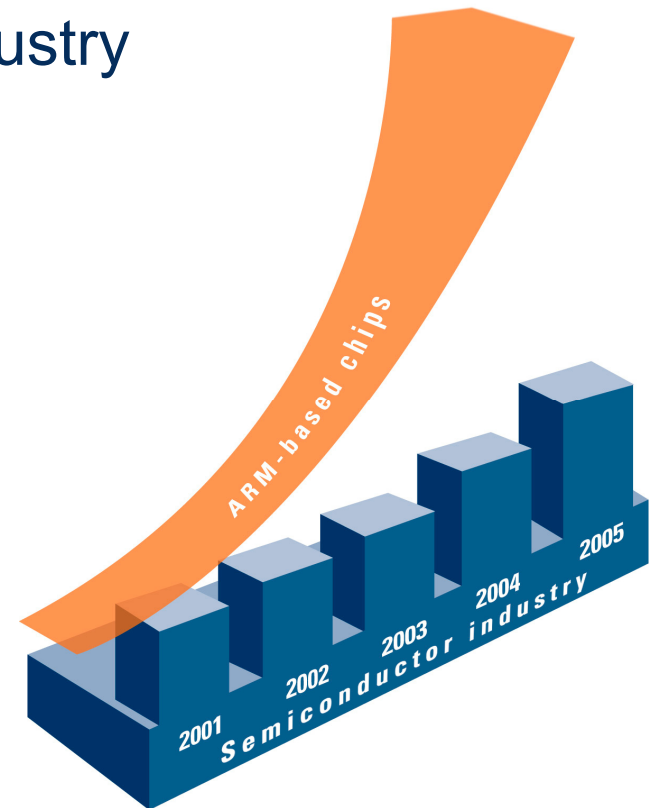
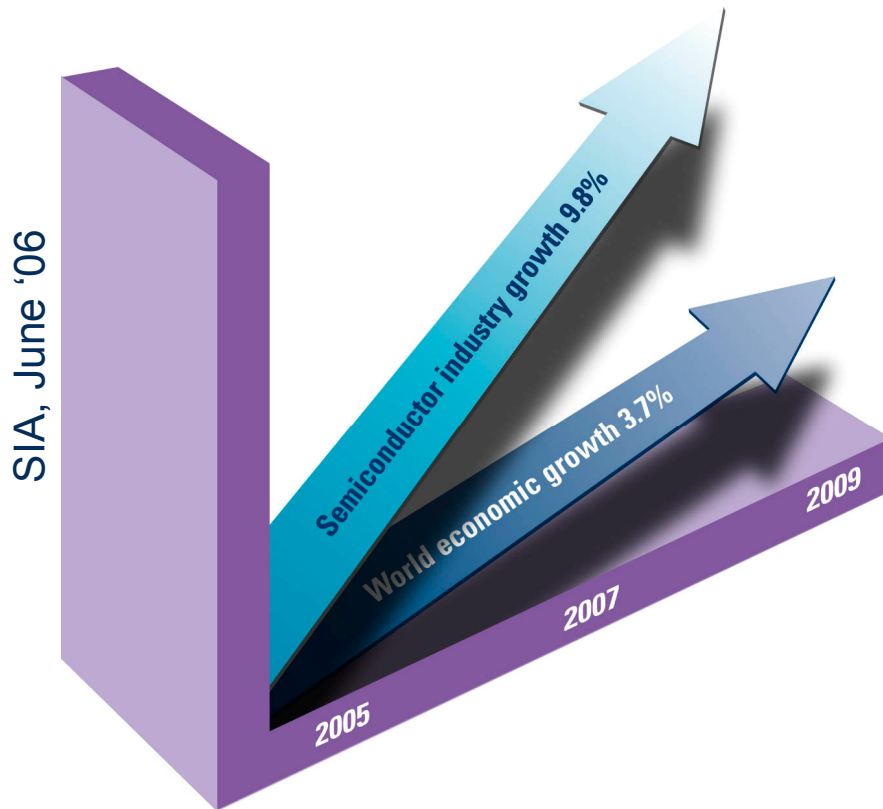
- High-performance devices run large amounts of complex software
- All needs to be developed and integrated into final device

Evolving to Meet the Changes



You Outpace the Rest of the World

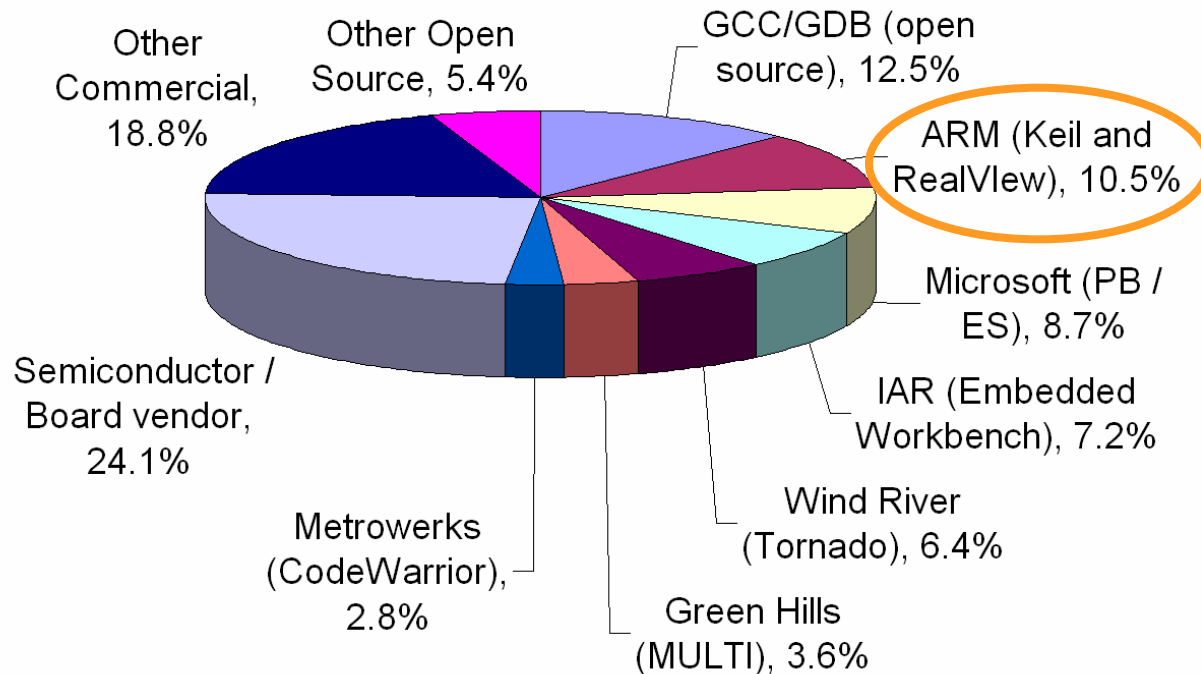
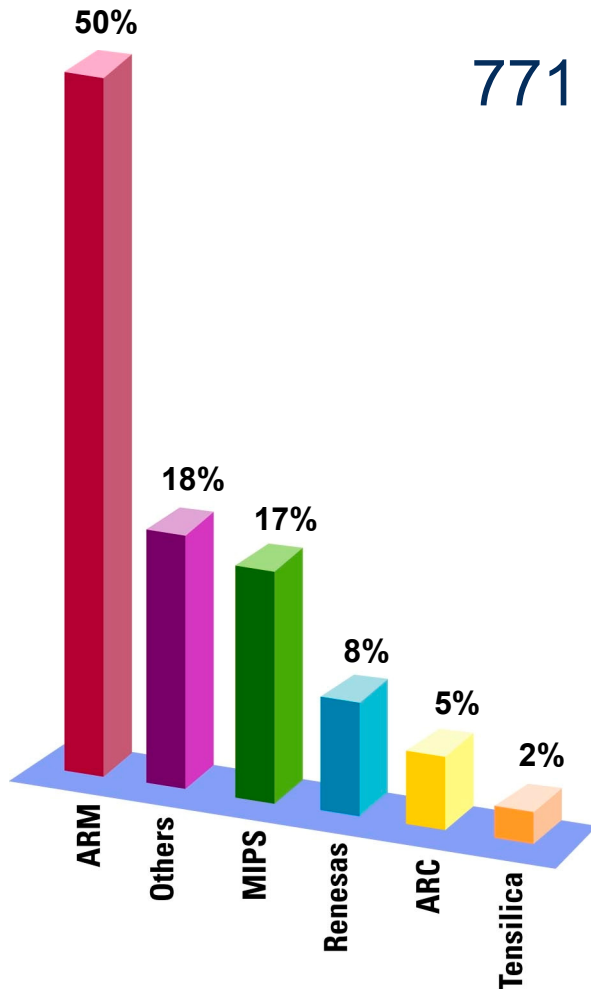
- Semiconductor industry outpaces the rest of the world economy
 - For 2006 and beyond
- ARM partnership outpaces rest of the semiconductor industry



ARM Partnership Outpaces Competition

771 new design starts in 2006

Source: Gartner Dataquest



*Total responses exceed 100% as some respondents are currently using more than one brand of RISC Microprocessor

Source: VDC Embedded Market Program 2005

ARM and Partnership Evolution

We are all evolving to meet the challenges

2001

- 77 processor licensees
 - 33 shipping 123 million/quarter
- Physical IP revenue \$6M/qtr
- ~\$10M R&D spend/quarter
- ~\$40B semiconductor market/quarter

2006

- 177 processor licensees
 - 69 shipping 572 million/quarter
- Physical IP revenue \$22M/qtr
- ~\$25M R&D spend/quarter
- ~\$60B semiconductor market/quarter

Integrated IP and tools delivers better products

ARM's Evolution in the Past Year

- Investing in new technology through acquisition and R&D
 - Two new companies
 - Falanx for 3D graphics
 - Keil MCU toolchain
 - Many new products and solutions including
 - Cortex™-A8, NEON™ and Cortex-R4
 - Swerve Java 3D engine and TrustZone® APIs
 - RealView® Development Suite 3.0
 - AMBA® Designer and Configurable AMBA 3 AXI™ PrimeCells®
- 51 new Partners in the ARM Connected Community

Improving Market Development

- Innovative products are not just about a technology solution
 - Business solution is required as well
 - Ecosystem must develop into a value chain
 - iPod + iTunes + content



- Content and user security requires a business solution

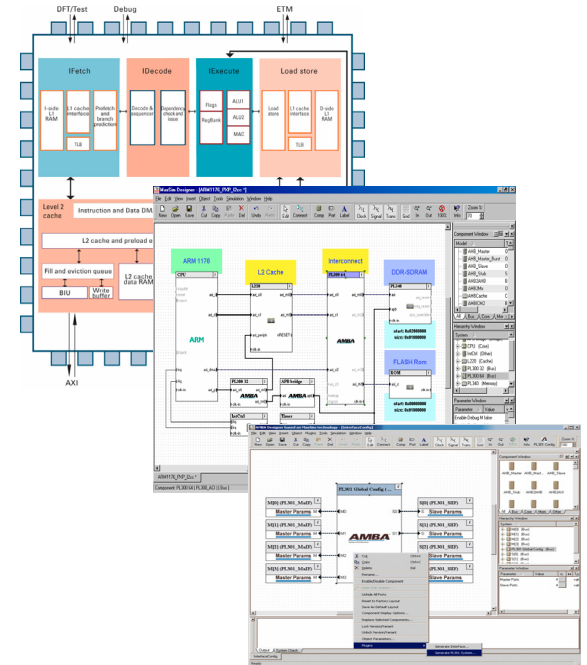
- SimLock, content protection, E-Commerce, privacy
- TrustZone is one component that links core, SoC, operating system and application software
- Complete solution requires ecosystem to form a value-chain
- ARM is working to put this in place

TrustZone®
Security Foundation by ARM®



Improving SoC Design

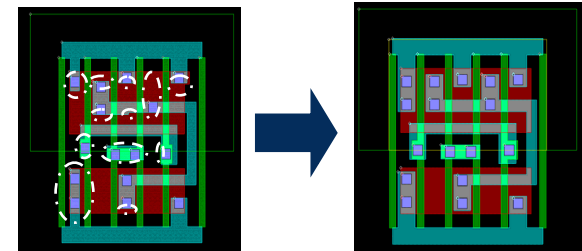
- Increased software performance requires correct hardware design and SoC architecture
 - Uni-core: Cortex-A8 with integrated NEON co-processor
 - Multi-processor: symmetric or asymmetric processing with MPCore™ technology
 - RealView® SoC Designer to get design right first time
 - AMBA 3 AXI and PL301 / AMBA Designer to implement and validate
 - Low latency memory system
 - CoreSight™ and RealView Debugger to support software development
 - Software validation using high speed virtual platforms



Improving Silicon Implementation

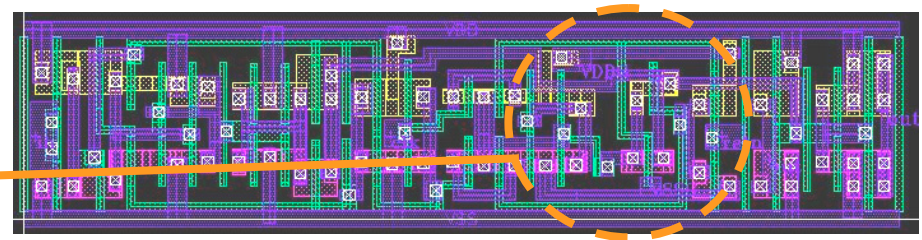
■ Yield-aware libraries

- Extensive knowledge of design rules
- DFM decks and proprietary software
- Highly robust DFM-aware libraries



■ Power Management Kits

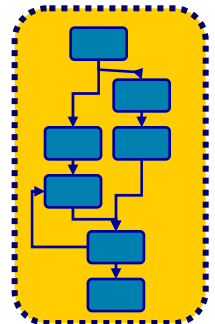
- Power gates (MT-CMOS)
- Retention flip-flops
- Back-bias support



■ Smoothing the path to implementation

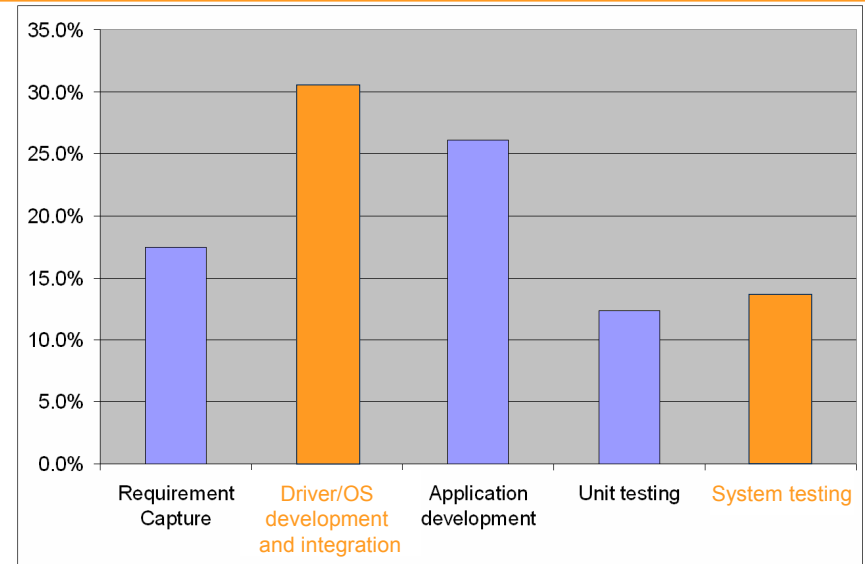
- Extensive collaborations with EDA vendors to create best practice power aware flows

cadence
MAGMA
SYNOPSYS®

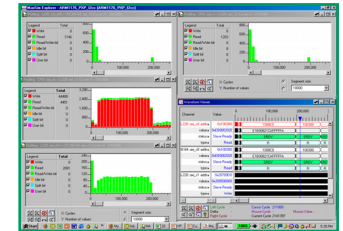
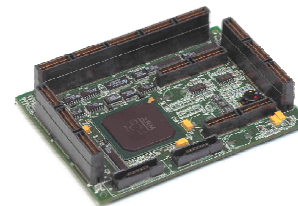
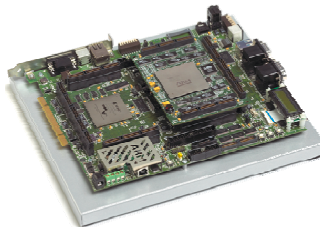
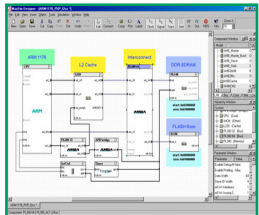


Improving Platform Integration

- Software and hardware integration is a time critical process
 - 50% of time is spent on OS porting and systems testing
 - Every bug is a delay to market
- ARM has technologies to parallel development and prevent delays
 - RealView SoC Designer system model for accurate OS-porting
 - Real Time System Model for fast application development
 - FPGAs and testchips for system integration
 - Single debug and compiler for all software development platforms
 - No need to change tool chain – removes massive risk



Embedded Software Market Intelligence Program
VDC, 2006



Improving Application Software

- Time-to-market and end-user experience are key
- Collaboration and standards enable reduced time-to-market
 - Sharing information and knowledge while protecting sensitive data
 - Do what you're the world's best at - collaborate in other areas

K H R O N O S
GROUP

OpenGL|ES™

OpenMAX

TrustZone®
Security Foundation by ARM®

- Best in class user experience enabled by software codecs or hardware acceleration
 - Working to the same standards for compatibility



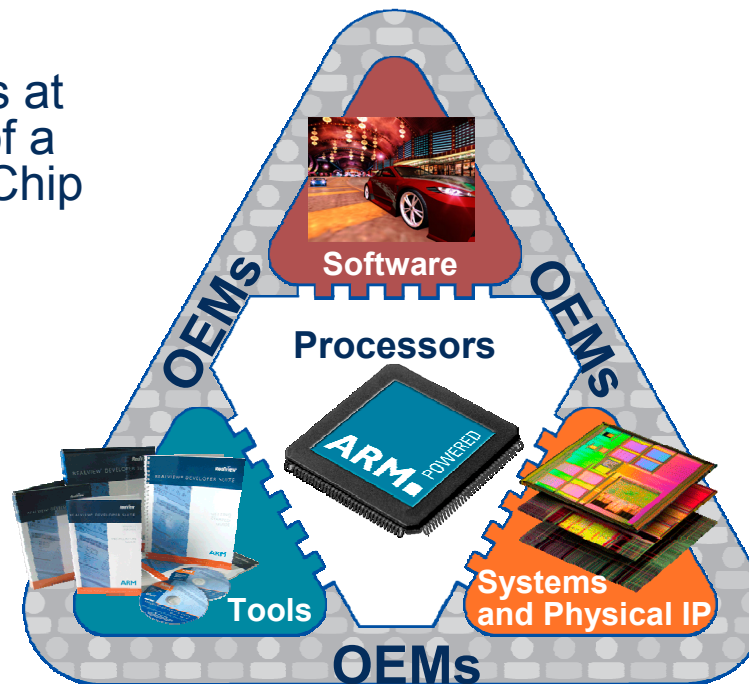
The ARM Strategy

The SoC must enable the device manufacturer to produce highly desirable products

The CPU is at the heart of a System on Chip

Software to bring the system alive and to maximise the performance of the system

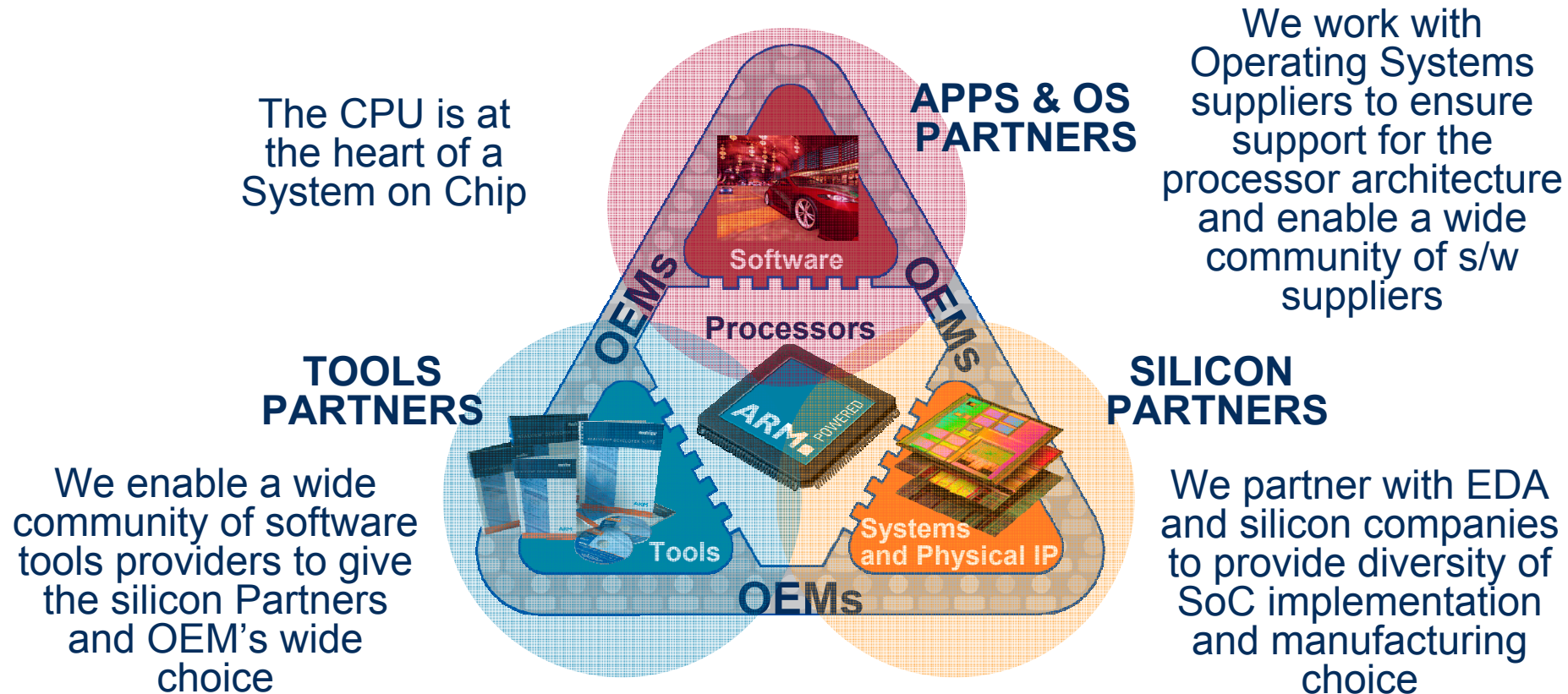
Software tools, ESL tools and models to target software onto the processors in the system



System on Chips require high performance Fabric along with high quality Physical IP

The ARM Strategy

Our strategy is to create a partnership with our customers and broader community of third parties, to enable the creation of end products more efficiently through ARM than from any other source



ARM Products From All Segments



Archos AV400
PMP ARM9



JVC Digital
Camcorder
ARM7TDMI®



Alfa Romeo Brera & 149
ARM9™ & ARM7™



Nintendo DS-Lite
ARM9 & ARM7



Sharp IT-32X2
LCD Television



Nokia High End Consumer
Phones



LEGO Mindstorm NXT
ARM7™ family-based MCU



Samsung Blu-Ray DVD
ARM926EJ-S™ + JTEK S/W



Samsung Camcorder
ARM9



Sony PEG-Vz90 Clie
Personal Digital Assistant



Toshiba 52HM84
52" DLP Television



Toshiba Gigabeat
ARM1136T2-S™